

Digital Clock User Guide

WiFi configurable LED clock - automatic time keeping, color profiles, temperature display, countdown, scoreboard and stopwatch modes.

This tri-fold guide summarizes first setup, daily use, advanced settings and troubleshooting. Scan the QR code for video help and the full guide.

Quick Access

Clock WiFi	DIGITAL-CLOCK-XXXXXX
Default password	12345678
Setup address	192.168.4.1
Guide website	ozmenclock.github.io/ozmen-clock-guide



Scan the QR code for the video guide, full HTML guide and PDF files.

Your phone may show a "no internet" warning. This is normal; the clock WiFi is only used to open the setup panel.

The setup page is not an internet website. First connect to the clock WiFi, then open 192.168.4.1 in your browser.

1. First Setup

- 1 Connect the clock to power and wait for the LED display to turn on.
- 2 Open the WiFi list on your phone or computer.
- 3 Connect to **DIGITAL-CLOCK-XXXXXX**.
- 4 Enter the default password: **12345678**.
- 5 Open **192.168.4.1** in your browser.

First setup screen

If the clock has not been set up before, the first setup screen appears. Choose language, country/city time zone, optional device WiFi password and home WiFi network.

The connection may briefly drop while saving the home WiFi network. Wait a few seconds and reopen 192.168.4.1 if needed.

After setup is complete, the normal settings menu opens. To change language or time zone later, open first setup again from the Tools tab.

2. Quick Start

- 1 Open the **WiFi** tab and select your home WiFi network.
- 2 Enter the WiFi password and press **Connect and Save**.
- 3 Select the correct time zone for your country.
- 4 If WiFi is connected, use **NTP** to update the time.
- 5 Open **Appearance** to adjust color and brightness.

For first use, set WiFi and time zone before adjusting appearance settings.

3. Main Tabs

Tab	Purpose
Appearance	Color, brightness and color profiles.
Time	12/24 hour, leading zero and dot LED settings.
Temp	Unit, offset and automatic temperature display.
Countdown	Enter time, start and stop countdown.
Scoreboard	Scores for 3 players or teams.
Stopwatch	Normal and lap stopwatch modes.
Night Light	Lower brightness during sleeping hours.
WiFi	Connection, time update and network status.
Tools	Password, first setup, reset, update and LED test.

Most settings are stored only after pressing the matching Save button. Live modes such as Scoreboard can change immediately.

4. Appearance Settings

Custom color

Selects the normal clock color from the color wheel and stores it with **Save Color**. The selected color is previewed on screen.

Color presets

Use one of the 12 preset colors for quick selection, such as red, green, blue, white and other common colors.

Brightness

Adjusts LED brightness. Higher brightness is easier to see during the day; lower brightness is better for bedrooms.

Digit and dot colors

Each digit and dot LED color can be selected separately.

Color profile

Selects effects such as Solid, Rainbow, Aurora, Lava, Prism and Fade. Low speed creates slower transitions; high speed makes animation more active.

5. Time and Dots

12/24 hour format

The clock can show time in 24-hour or 12-hour format.

Dot LED behavior

Dots can blink or stay always on.

NTP time update

When connected to home WiFi, internet time is received and stored in the clock. If the region is selected correctly, daylight saving follows that time zone.

Update Time

Sends phone or computer time to the clock. This is useful when internet time is not available.

If internet is not available, the clock continues using its stored RTC time.

The RTC module helps the clock keep time even after power loss. Small drift during long offline use is normal; NTP corrects it again when WiFi is available.

6. Temperature Display

- If a sensor is installed, room temperature can be displayed.
- **Show Temperature** briefly shows the measured temperature on the LED display.
- Celsius or Fahrenheit can be selected.
- Auto display controls how often and how long temperature appears.
- Offset corrects readings if they are slightly high or low.

If the model does not include a temperature sensor, temperature may stay hidden while the clock functions continue normally.

When automatic temperature display is enabled, the clock briefly shows temperature at the selected interval, then returns to the normal time display.

7. Night Light

Automatically lowers display brightness during sleeping hours.

- **Start:** Hour and minute when night mode begins.
- **End:** Hour and minute when normal brightness returns.
- **Night brightness:** Lower brightness used during this period.

Low brightness is recommended for bedrooms and dark rooms.

Night Light only lowers brightness during the selected hours; color, hour format and other settings remain unchanged. Normal brightness returns after the end time.

8. Countdown

- 1 Enter hours, minutes and seconds.
- 2 Use **Show Entered Time** to preview before starting.
- 3 Press **Start**.
- 4 Use **Pause** and **Resume** to control it.
- 5 Use **Reset** to return to the saved time.

Countdown LED color can be selected separately. This mode is useful for events, sports, training and shop displays.

While countdown is active, the clock leaves the normal time display. When finished, or when **Show Clock** is used, it returns to normal clock mode.

9. Scoreboard and Stopwatch

Scoreboard is used for 3 players or teams. Left, middle and right scores can have different colors and change instantly with + / - buttons.

Stopwatch can run in normal or lap mode. In lap mode, the lap count increases when the selected lap time is reached.

Score colors can be saved; score values are used live during operation. Stopwatch color can also be selected so the mode is easy to identify on the LED display.

10. WiFi and NTP

- The WiFi tab shows network, MAC address, IP, signal and quality.
- **Scan WiFi Networks** lists nearby networks.
- **Connect and Save** stores home WiFi details.
- UTC is set according to the region selected during first setup.
- **NTP** can update the clock from internet time.

A brief disconnection may happen while connecting to WiFi. This is normal.

When the clock joins the home WiFi network, its local IP address appears in the WiFi tab and can be used from the same network.

If the home WiFi is not available, the clock keeps its own **DIGITAL-CLOCK** network active so the user can reconnect and enter new WiFi details.

11. Tools and Confirmations

- Device WiFi password can be changed with current password confirmation.
- The first setup screen can be opened again.
- Firmware update page is opened from the Tools tab.
- LED test checks all LEDs.
- Red actions ask for a centered confirmation dialog.

Factory reset should be used only when setup is wrong or the device WiFi password is forgotten. WiFi details must be entered again after reset.

For firmware update, upload only the compiled **.bin** file. Do not disconnect power until the update is complete.

12. Troubleshooting

Problem	Check
192.168.4.1 does not open	Make sure you are connected to the clock WiFi network.
WiFi does not connect	Check password, 2.4 GHz network and router coverage.
Wrong time	Use NTP while WiFi is connected or press Update Time.
Password forgotten	Use factory reset. Default password returns to 12345678.

For connection issues, first check that you are connected to the clock WiFi, then confirm the IP address in the browser. If the home WiFi password changed, save it again from the WiFi tab.

Use and safety: Keep the device in a dry place. Do not power off during firmware update. Loose power cables or weak adapters may cause unstable LEDs or WiFi.